

Photon-budget-dependent estimation strategy for nitrogen-vacancy Ramsey magnetometry

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1

(Dated: September 14, 2026)

Estimating a static magnetic field from a Ramsey fringe is usually discussed as a problem of photon shot noise, as if the inference itself were free. Working with a real nitrogen-vacancy (NV) diamond data set that spans a $128\times$ range of photon budget (5×10^3 to 640×10^3 repetitions per delay point, 300-point sweep, 40 field settings), we show that the sensitivity-optimal *inference strategy* changes with the photon budget (about 30 to 4×10^3 detected photons per delay point). Per-trace maximum-likelihood fitting is essentially efficient against its own Cramér–Rao bound at every budget (743 nT at 5k repetitions against a free-envelope bound of 744 nT), yet is a factor 1.63 worse than the pooled estimator whose bound is lower because the instrument’s shared nuisance structure, one relaxation envelope and one phase frame for a whole field sweep, is estimated once for the whole sweep rather than once per trace. Both estimators obey a two-term law $\delta B(r) = \sqrt{A^2(5000/r) + b^2}$ with $A = 785.3$ nT, $b = 27.8$ nT for per-trace fitting and $A = 456.0$ nT, $b = 57.6$ nT when the pooling is restricted to field settings that are individually constrainable, so that pooling is the better strategy essentially throughout the measured ladder. Pooling the full sweep instead — including τ records too short for the field to be identifiable from a single trace — degrades the shared nuisance and produces an apparent ~ 150 nT floor that advances the crossover to $r^* \approx 1 \times 10^5$ repetitions. The choice of *which* settings enter the pooled session is therefore as important as the choice of pooling itself, and it is what a synthetic control with exact ground truth isolates. The criterion is pre-computable — a single-trace bound smaller than the field step — so the pooled session can be chosen before the measurement. An amortized neural estimator trained once across the whole budget range, without being told the instrument model or which parameters are shared, stays within 2–21 percent of the best per-budget classical estimator at every budget — a consistency that no single classical estimator achieves — although it is never itself the best. The practical rule that follows is not “pool then stop sharing” but “pool over settings you can also constrain individually”: an instrument’s sweep often contains records that a single trace cannot identify, and those records corrupt the very shared nuisance that pooling is meant to estimate. Both the law and the crossover are conditional: they hold for an identifiability-filtered session, and for data whose blind FFT+Rife initialisation lands inside the refinement window — a condition we verify on these traces (median initialisation error 1.1×10^3 nT against a $\pm 3.2 \times 10^3$ nT window) but which an analysis of other data must re-check.

I. INTRODUCTION

Magnetometry with nitrogen-vacancy (NV) centres in diamond has become a workhorse of quantum sensing [1–5]. In the Ramsey protocol, a superposition of the NV spin states accumulates a field-dependent phase $\phi = \gamma B \tau$ over a free precession time τ , and the resulting population oscillation is read out optically. Repeating over a grid of delays yields a damped fringe whose oscillation frequency fixes B . The sensitivity of such a measurement is conventionally written $\eta = \delta B \sqrt{t_{\text{total}}}$, where δB is the field uncertainty and t_{total} the accumulation time of the data used [5].

Almost all of the sensitivity discussion in the literature concerns the *hardware and protocol* side of that expression: contrast, collection efficiency, coherence time [5, 6], or the choice of which delay times to measure [7]. The inference side, namely how close a practical estimator actually comes to the information-theoretic bound for a given finite data set, is rarely the headline, even though it sets the achievable sensitivity whenever photons are scarce. Learned estimators for NV data have appeared recently [8–12], but they are usually benchmarked on sim-

ulations or against weakly tuned classical baselines, and the simulation-to-reality gap is repeatedly identified as the central difficulty [9, 11].

In this work we ask the estimator-side question directly, on real data, with a fixed measurement protocol. We use a DC Ramsey data set in which the same 40 field settings are measured on 8 sheets that differ only in the number of repetitions per delay point (5×10^3 to 640×10^3 , thus a $128\times$ photon range), always with the full 300-point delay sweep. This lets us compare estimators at *matched photon budget*. We deliberately do not study sparse or adaptive delay sampling: the sweep is fixed and only the photon budget varies.

Our main finding is that the optimal inference strategy depends on the photon budget, and that *which* measurements enter the inference matters as much as how they are combined. A per-trace fit, with six free parameters per fringe, is efficient against its own Cramér–Rao bound, but that bound is inflated by the need to determine the fringe’s nuisance parameters from a single trace; pooling an instrument’s shared nuisance structure across the field settings of a sweep moves the bound down by a factor close to two. The pooling is only as good as the

settings it is built from, however. A sweep typically contains records whose field a single trace cannot identify at all, and those records contribute little information about the field but a great deal of noise to the shared nuisance, degrading every other setting in the session. Quantifying that effect, and giving an objective criterion for which settings may be pooled, is the content of this paper.

II. EXPERIMENT AND DATA

The data set consists of DC Ramsey measurements on an NV ensemble, provided as 8 sheets of a 300×40 array. Rows are delay times $\tau_i = 20i$ with $i = 1, \dots, 300$ and a step of 20 ns, i.e. 0.02 to $6.0 \mu\text{s}$. Columns are applied fields $B_n = 1071.4n$ with $n = 1, \dots, 40$, i.e. 1071.4 nT steps covering 1.07 to $42.9 \mu\text{T}$. Each sheet corresponds to a different number of laser–microwave repetitions per delay point, $r \in \{5, 10, 20, 40, 80, 160, 320, 640\} \times 10^3$, which at this readout corresponds to approximately $r/150$ detected photons per delay point (roughly 30 to 4×10^3 photons per point across the ladder); the signal-to-noise ratio scales with r as expected for shot noise. All evaluations in this paper use the real traces; no synthetic data appears in any reported performance number. Synthetic traces are used only to train the neural estimator described in Sec. IV F and, with *exact* ground truth, in the control experiment of Sec. IV C.

We use the standard model for a Ramsey fringe with a stretched-exponential envelope,

$$s(\tau) = C + A e^{-(\tau/T_2^*)^p} \cos[\gamma B(\tau + \tau_{\text{off}}) + \phi_0], \quad (1)$$

with $\gamma = 2\pi \times 28 \times 10^{-6} \text{ rad}/(\mu\text{s nT})$. Two instrument conventions are adopted from prior work on the same apparatus and are treated as known throughout: a timing offset $\tau_{\text{off}} = 18.6 \pm 2.6 \text{ ns}$ [14], and the global phase frame of a session, i.e. the instrument phase ϕ_0 is common to all field settings of one sweep, as we verify below.

Noise-model validation. The bounds and efficiencies below assume that the per-point noise is additive, stationary and uncorrelated. We verified this on the real traces: the lag-1 autocorrelation of the residuals of the per-trace fits, averaged over columns 7–40, is -0.010 , -0.010 and $+0.004$ at 5k, 40k and 640k repetitions respectively, consistent with the ± 0.058 expected for white noise of 300 points. The Fisher information used for the bounds is therefore appropriate for these data, and a residual structure (drift, correlated readout noise) is not inflating the apparent estimator efficiency.

Noise. A robust per-point noise estimate (standard deviation of the three-point second difference, divided by $\sqrt{6}$) gives 0.2079, 0.1500, 0.1072, 0.0745, 0.0533, 0.0370, 0.0263, 0.0183 for the eight sheets, matching the shot-noise law $\sigma(r) = 0.207\sqrt{5000/r}$ to within 0.2–3.5%. We use that law wherever a noise level is needed.

Envelope population. Fitting Eq. (1) to the 40 columns of the highest-budget sheet gives $A = 0.127 \pm 0.005$, $C =$

0.889 ± 0.006 , $T_2^* = 5.404 \pm 0.294 \mu\text{s}$, $p = 2.014 \pm 0.423$ and $\phi_0 = -0.044 \pm 0.123 \text{ rad}$ (Fig. 4). The phase scatter is consistent with a common instrument phase: the circular concentration of ϕ_0 across columns is 0.955 at 5k, 0.971 at 10k and 0.993–0.992 at 320k and 640k repetitions. The relaxation time, by contrast, genuinely varies by $\pm 5\%$ across the field range, a fact that will matter in Sec. IV C.

Ground truth and its systematic. The applied field is the nominal value B_n . Fitting the highest-budget sheet column by column, the fitted fields deviate from the nominal values by 44 nT (median absolute deviation, columns 7–40; range -107 to $+139$ nT). This is a property of the magnet and its calibration rather than of any estimator, and it is why all comparisons below are made against B_n while its ~ 50 nT scale is stated explicitly: at the two highest budgets the per-trace statistical error approaches this scale, and we mark that boundary in the figures.

III. MODELS, BOUNDS AND ESTIMATORS

A. Estimation task

We estimate an unknown field from one full 300-point trace. To exclude any possibility of scoring an estimator by information contained in the experimental set point, the task is *blind*: the estimator is given $B \in [0.5, 45] \mu\text{T}$ and nothing else about the field. All classical estimators are initialised identically from a blind FFT+Rife [13] estimate of the same trace, which is standard practice for fringe fitting, so that differences between estimators reflect the refinement step alone. We report results for columns 7–40 ($B \geq 7.5 \mu\text{T}$, at least ~ 1.3 oscillations within the sweep) as the primary region. Columns 1–6 are treated separately: those sub-cycle records are not identifiable per trace at low budget by any method we tried, and their calibrated treatment belongs to companion work [14].

B. Cramér–Rao bounds and why pooling moves them

For a fixed sweep, three bounds are relevant. If the envelope (A, C, T_2^*, p, ϕ_0) is known, the field bound σ_B^{known} follows from the Fisher information $\mathcal{I}_B = \sum_i (\partial s_i / \partial B)^2 / \sigma^2$. If the envelope is free, the bound inflates to σ_B^{free} , because six parameters must be determined from the same trace. If instead the envelope is *shared* across the 40 field settings of a sweep while each column keeps its own field, the relevant bound σ_B^{joint} follows from the joint Fisher matrix: the nuisance cost is paid once for the whole session rather than once per trace. For the measured trace geometry at 5k repetitions these are 744 nT (free, per trace) and 419 nT (joint, 40 columns). The ratio $\sigma_B^{\text{free}} / \sigma_B^{\text{joint}} \approx 1.8$ is set by the fringe geometry, namely the number of oscillations, the decay and the sampling, and is essentially independent of the

photon budget (Fig. 2, left panel). Because the ratio is set by the trace geometry, it neither grows nor shrinks with the number of repetitions — which is why the strategy choice below is decided by the competition between this gap and the bias floor, not by the gap alone.

The scientific content of this paper is what happens when these bounds are confronted with real estimators on real data, including the fact that the shared-envelope model is not exactly true for a real instrument.

C. Estimators

Per-trace fitting (1m). Levenberg–Marquardt least squares with multi-start over the initial phase, fitting all six parameters of Eq. (1) from the FFT estimate, with nine starts inside the blind window. Raising the number of starts to 25 changes the fitted fields by a median of 0.03 nT and by at most 39 nT at the lowest budget — an order of magnitude below the 743 nT error obtained there — so the baseline is not start-limited at the precision that matters here.

Session-pooled fitting (pool). A single least-squares problem over all 40 columns of a sweep: one (A, C, T_2^*, p, ϕ_0) shared by every column and one field per column. This is the classical estimator that exploits the shared instrument structure, and its bound is σ_B^{joint} .

Partial pooling (part). As above, but each column keeps its own T_2^* while A , C , p and ϕ_0 remain shared. This structure follows the measured instrument: the phase frame is genuinely common, the relaxation time is not.

Grid Bayes (gb). A third, non-least-squares classical estimator, included so that the comparison is not restricted to LM variants: a marginal posterior over B on a 161-point grid spanning the same $\pm 3.2 \times 10^3$ nT window, with A and C profiled by linear least squares and (T_2^*, p, ϕ_0) marginalised on a coarse grid ($3 \times 3 \times 9$ nodes); the reported estimate is the posterior mean.

Amortized network. See Sec. IV F.

D. Metrics

We report the root-mean-square error over the 40 real traces of a sheet, which is a σ -equivalent statistic, together with the signed bias and a percentile bootstrap confidence interval over columns. A quoted median is related to the Gaussian-equivalent σ by the usual factor 0.674. Each estimator is compared with its *own* bound: the free-envelope bound for per-trace fitting and the joint bound for the pooled estimators. The sensitivity uses a declared per-point overhead, $t_{\text{total}}(r) = r \sum_i (\tau_i + t_{\text{ovh}})$, for $t_{\text{ovh}} \in \{0.5, 1, 2\}$ μs . All reported fractions and rankings are overhead-independent, because every estimator is charged the same session time.

IV. RESULTS

A. Both estimators are efficient, but against different bounds

Figure 1 and Table I give the central measurement. In the primary region, per-trace fitting attains 743 nT at 5k repetitions against its own free-envelope bound of 744 nT — an efficiency (RMSE divided by that bound) between 1.0 and 1.3 for $r \leq 320\text{k}$ and 0.90 at 640k. The classical fitter is therefore essentially not improvable by a better optimiser. The bound is evaluated with each column’s own fitted envelope and the measured noise law; the 10% sub-unity value at 640k is within the uncertainty of those inputs and is not interpreted as super-efficiency. Session pooling attains 455 nT against a joint bound of 419 nT, also efficient, and is better than per-trace fitting by a factor 1.63 at 5k repetitions (1.97 at 10k). The marginal-posterior estimator sits between the two and changes places with per-trace fitting: it is the better of the two per-trace analyses at the four lowest budgets (669 against 743 nT at 5k, 237 against 262 nT at 40k), where marginalising the envelope pays, and it loses above 80k (174 against 59 nT at 640k) because its residual error is set by the coarseness of the nuisance grid rather than by photons. Pooling therefore does not recover lost efficiency; it changes the bound. This is the first result: at low photon budget the achievable precision is limited by the identifiability of the field jointly with the instrument’s nuisance parameters, not by the optimiser.

Paired Wilcoxon signed-rank tests over the 34 real columns of the primary region confirm the ordering: the pooled-versus-per-trace contrast has $p = 3.7 \times 10^{-4}$ at 10k and 3.2×10^{-5} at 40k repetitions, while at 5k it is marginal ($p = 9.8 \times 10^{-2}$) because the per-trace error distribution has a heavy tail there; the pooled estimator remains ahead at 80k and 160k ($p = 8.1 \times 10^{-3}$ and 8.1×10^{-3}). Above that the two become statistically indistinguishable ($p = 1.1 \times 10^{-1}$ at 320k and 3.2×10^{-1} at 640k). We therefore present the ladder as a curve described by a two-parameter law rather than as eight independent significance claims, with the two lowest budgets as the pre-declared primary contrast.

B. The crossover law

Table I shows that with the primary-region session the pooled estimator remains ahead of per-trace fitting at every budget up to 320k repetitions and falls behind by 23% at 640k (73 against 59 nT); the two bias curves in the lower panel of Fig. 1 stay comparable over the whole ladder.

The whole ladder is described by a two-term law,

$$\delta B(r) = \sqrt{A^2 (5000/r) + b^2}, \quad (2)$$

in which A is the photon-limited coefficient and b a budget-independent floor. Fitting the measured ladders

TABLE I. Field RMSE (nT) over the real columns 7–40 as a function of repetitions per delay point: per-trace LM, session pooling (shared envelope), partial pooling (per-column T_2^*) and the amortized session-conditioned network. The classical columns are generated from `results/analysis.v2.json` and the network column from `results/blind_res_nets.json`, both by `src/make_numbers.py`; no value in this table was entered by hand.

r	lm	pool	part	network
5k	743	455	484	504
10k	629	319	324	387
20k	383	279	278	289
40k	262	147	148	172
80k	195	114	117	139
160k	165	98	98	113
320k	109	86	86	88
640k	59	73	73	66

TABLE II. Pooled estimation under two session definitions and three sharing patterns, real data, columns 7–40 scored in every case. “Full sweep” pools all 40 settings, including the sub-cycle records (columns 1–6) whose field is not identifiable from a single trace at low budget; “primary region” pools only the 34 settings that are.

pooled session	A (nT)	b (nT)	r^*
primary region, all shared	456.0	57.6	6.2×10^5
primary region, T_2^* per column	474.8	53.2	6.9×10^5
full 40-setting sweep	416.8	156.2	9.1×10^4

gives $A = 785.3$ nT and $b = 27.8$ nT for per-trace fitting (fit rms 32.6 nT) and $A = 456.0$ nT and $b = 57.6$ nT for pooled fitting (fit rms 18.8 nT). The photon-limited coefficients differ by $(A_{\text{lm}}/A_{\text{pool}})^2 = 2.97$: pooling is worth a factor $\sqrt{2.97}$ in precision while photons are scarce, and with $b_{\text{pool}} = 57.6$ nT the two curves cross only at $r^* \approx 6.2 \times 10^5$ repetitions. Expressed operationally: to reach the precision that pooled inference achieves at 5×10^3 repetitions, per-trace fitting needs 2.97 times more repetitions, and throughout the measured range up to 640k the pooled estimator is never the worse choice by more than the ~ 50 nT per-column systematic discussed above.

C. What sets the floor: the session definition

The two-term law above was fitted to a pooled session built from the 34 field settings of the primary region. That choice is not innocent, and it is the one place where our earlier analysis changed its mind: an equally natural protocol — pooling over the *full* 40-setting sweep, which is what an experimenter actually has — gives a markedly different law. The comparison is in Table II.

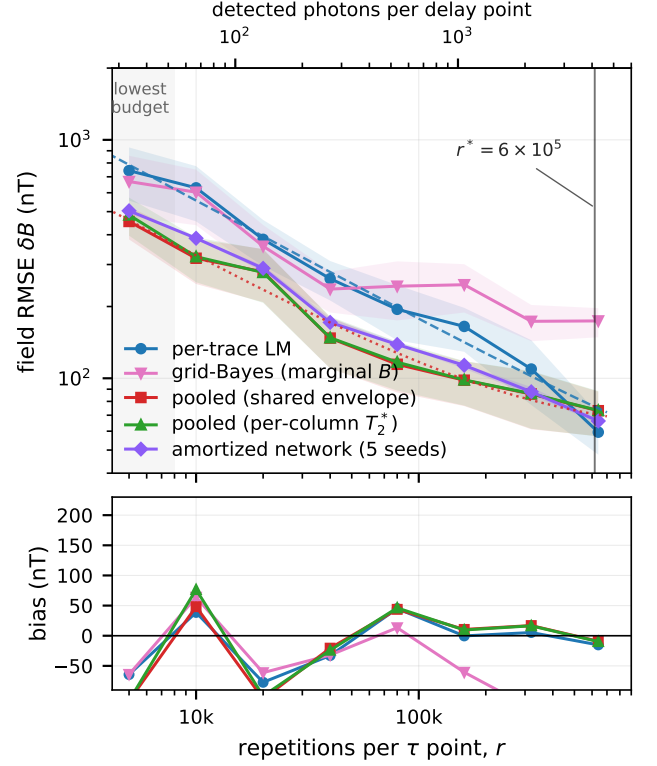


FIG. 1. Real data, columns 7–40. Top: field RMSE against repetitions per delay point for the five estimators (three least-squares variants, the marginal-posterior estimator, and the network). Shaded bands are *variability*, not confidence intervals: for the classical estimators they span a 2000-resample percentile bootstrap over the 34 columns, which quantifies how consistently an ordering holds across field settings within this single session and not a sampling uncertainty of the method; for the network the band is the interquartile spread across the five training seeds. Dashed and dotted curves are the two-term law of Eq. (2) fitted to the per-trace and pooled ladders; the vertical line is the crossover r^* . Bottom: signed mean error of the same estimators on the same traces (symmetric-log scale). The crossover in the top panel is the point where the pooled estimators’ growing bias, visible in the bottom panel, overtakes their bound advantage. The shaded strip marks the lowest measured budget.

Pooling the full sweep produces a bias floor of 156.2 nT against 57.6 nT for the primary-region session, and advances the crossover from 6.2×10^5 to 9.1×10^4 repetitions — a factor of seven. The mechanism is visible directly: the sub-cycle records contribute little information about the field but a great deal of noise to the shared envelope, and a nuisance estimated from them is worse for every other setting in the sweep. Relaxing the most inhomogeneous shared parameter (T_2^* , which varies by $\sigma = 0.294 \mu\text{s}$, i.e. 5% of its mean, over the field range) recovers about 40% of the difference, but not all of it: freeing T_2^* per column in the full-sweep session moves the floor from 156.2 nT to 92.0 nT and the crossover

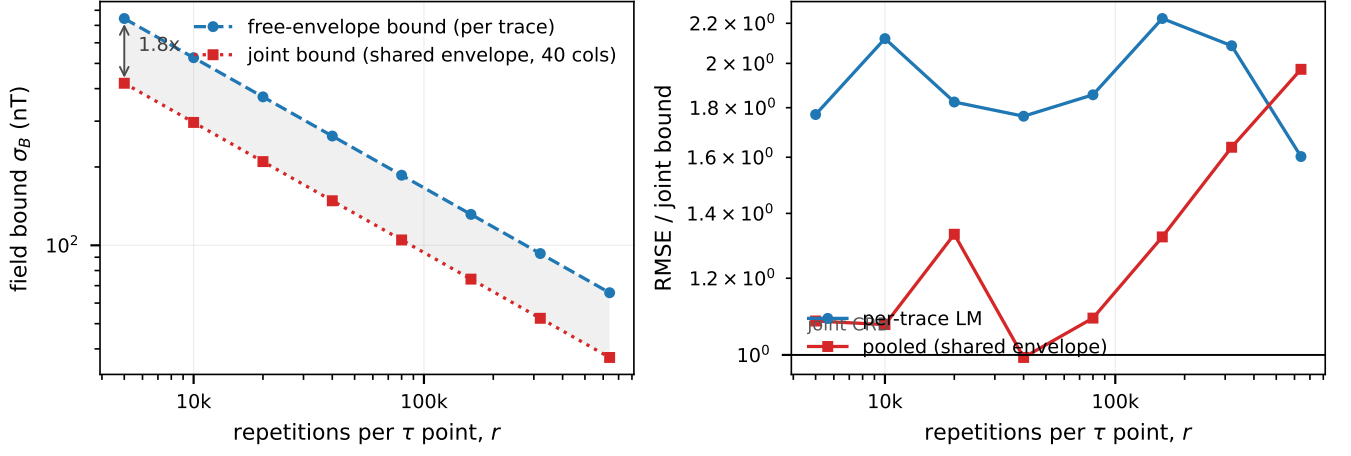


FIG. 2. Left: the two Cramér-Rao bounds for the measured trace geometry, median over columns 7–40 — the free-envelope bound that a per-trace fit must respect, and the joint bound available when the envelope is shared across the 40 field settings. The shaded gap is the factor ≈ 1.8 that pooling is worth. Right: the measured RMSE of each estimator divided by the *same* (joint) bound, so that neither can appear efficient by being normalised to its own reference. The per-trace estimator lies $1.6\text{--}2.2\times$ above the joint bound at every budget (the ratio has no clear trend with photon budget); the pooled estimator lies within 10% of the joint bound at most budgets, with excursions to $1.3\times$ at 20k and 160k, and rises to $2.0\times$ at the highest budget, where the small residual floor of Eq. (2) takes over.

from 9.1×10^4 to 2.4×10^5 repetitions, so part of the penalty is carried by the other shared parameters and by the unidentifiable settings themselves.

This is the quantitative version of a caution that applies to any session-pooled analysis: *pool over settings whose parameters you could also estimate one at a time*, or the pool will be dragged by the records that carry the least information.

The synthetic control of Fig. 3 tests the corresponding conclusion with exact ground truth. It is a *full-sharing* control — one shared envelope for a session of randomly drawn field settings, no measurement systematics — and it reproduces the full-sharing behaviour rather than the primary-region one: with per-trace 520.7 against pooled 374.5 nT at 5k and 51.7 against 217.8 nT at 640k (medians, as this control reports), its fitted law has a pooled floor of $\sim 0.4 \mu\text{T}$ and a crossover near 3×10^4 repetitions. Since the synthetic session contains no unidentifiable settings and no instrument drift, that floor is entirely a property of sharing one envelope across heterogeneous settings. The primary-region result is therefore a statement about which settings may enter the pool, and the full-sharing control is the limiting case it avoids.

D. An objective identifiability criterion, and the add-back test

The primary region was defined in Sec. III A qualitatively (settings whose field completes at least ~ 1.3 oscillations within the sweep). For the session-definition claim to be more than an ad hoc exclusion it needs an objective criterion, and there is a natural one: a setting

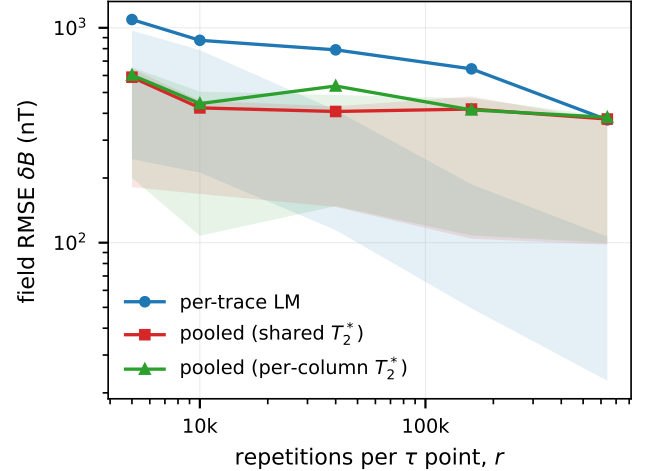


FIG. 3. Synthetic control with exact ground truth and the measured envelope population: field RMSE against budget for per-trace fitting, full pooling and partial pooling, with interquartile bands. With no measurement systematics and no unidentifiable settings present, the pooled estimator still acquires a floor near $0.4 \mu\text{T}$ and is overtaken by per-trace fitting above $\sim 3 \times 10^4$ repetitions, which isolates the floor as a property of sharing one envelope across heterogeneous settings.

is individually identifiable if its *single-trace* free-envelope bound is smaller than one field step, i.e. if a single trace can at least place the field in the right bin. Evaluating that bound at the lowest measured budget (5k repetitions) marks columns 1–7 as unidentifiable (their bounds

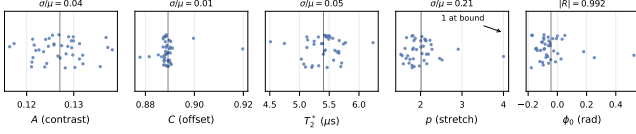


FIG. 4. Relaxation envelope measured across the 40 field settings from the highest-budget sheet (strip plots, one dot per field setting; the vertical line is the mean). Each panel gives the scatter: the contrast, offset and relaxation time vary by a few per cent, the stretch exponent varies by 21% and includes one setting at the numerical bound $p = 4$, and the phase is essentially common ($|R| = 0.992$ is the circular mean resultant length, the analogue of a correlation coefficient for angles). These scatter values are the model-mismatch budget that the pooled estimators of Fig. 1 must absorb.

TABLE III. Add-back test: pooled session grown from the 34 identifiable settings by adding the weak ones in order of decreasing identifiability. A , b are the coefficients of Eq. (2); r^* is the resulting crossover with per-trace fitting. The last two rows fail to produce a usable joint fit.

settings	A (nT)	b (nT)	r^*
34 (identifiable)	456	58	6.2×10^5
35 (+ col. 6)	454	59	5.8×10^5
36 (+ col. 5)	447	62	5.4×10^5
37 (+ col. 4)	441	63	5.3×10^5
38 (+ col. 3)	436	67	4.8×10^5
39 (+ col. 2)	2339	0	—
40 (+ col. 1)	2396	0	—

range from 4.1×10^3 to 4.9×10^4 nT, i.e. 4 to 45 field steps of 1071.4 nT) and columns 8–40 as identifiable, so the criterion selects 33 of the 40 settings — within one column of the 34 used above. Using the strict 33-setting session changes the law modestly — $A = 459.4$ nT (+0.7%), $b = 55.4$ nT (−3.8%) and $r^* = 6.6 \times 10^5$ (+7%) against the 34-setting values — so the conclusion does not hinge on the exact cutoff, though the crossover does move by more than the coefficients because it divides by b^2 .

Adding the weak settings back one at a time (Table III) degrades the law continuously rather than abruptly: each of the four best-identified weak settings costs a few per cent, and the two least identifiable ones (columns 2 and 1, single-trace bounds of 45 and 39 field steps) destroy the joint fit outright, driving A to $\sim 2.4 \times 10^3$ nT and b to zero. The failure is not gradual at the end: those settings contribute essentially no information about the field but a great deal of likelihood volume along the shared-noise directions.

E. Sensitivity

With the declared accounting $t_{\text{total}}(r) = r \sum_i (\tau_i + t_{\text{ovh}})$, the sensitivity $\eta = \delta B \sqrt{t_{\text{total}}}$ at $t_{\text{ovh}} = 1 \mu\text{s}$ reads

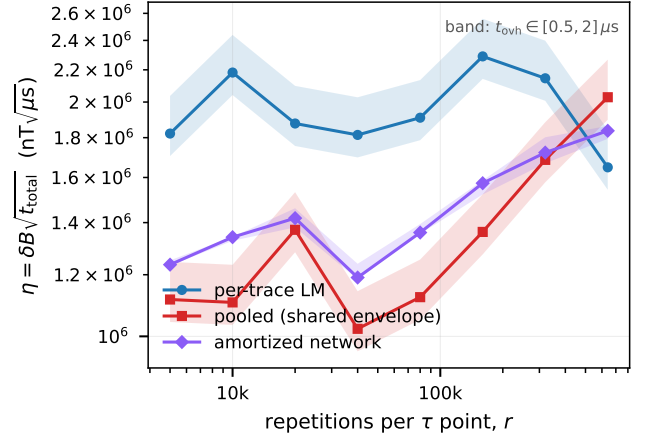


FIG. 5. Sensitivity $\eta = \delta B \sqrt{t_{\text{total}}}$ against repetitions per delay point for per-trace and pooled inference, and for the amortized network; the shaded bands show the per-point overhead t_{ovh} varied over $0.5\text{--}2 \mu\text{s}$, which rescales η for every estimator alike and therefore leaves the ranking unchanged.

$\eta_{\text{pool}} = 1.12 \times 10^6$ and $\eta_{\text{lm}} = 1.82 \times 10^6$ nT $\sqrt{\mu\text{s}}$ at 5k repetitions (Fig. 5), and $\eta_{\text{pool}} = 2.03 \times 10^6$ versus $\eta_{\text{lm}} = 1.65 \times 10^6$ at 640k. The ranking inverts at the same crossover as the field error, and it is unchanged under $t_{\text{ovh}} \in \{0.5, 1, 2\} \mu\text{s}$, because every estimator is charged the same session time: the overhead rescales η without reordering the estimators. This is the practical form of our result. For a fixed sweep and a fixed photon budget there is a best inference strategy, and it changes at r^* .

F. An amortized estimator that chooses its own pooling

The two-term law tells an experimenter *which* strategy to use, but it requires the instrument’s envelope statistics and the photon budget to be known in advance. We therefore trained a neural estimator that spans the entire budget range with a single set of weights and asked how close it comes to the best classical strategy at each budget. The network reads all 40 traces of a session and, for each column, predicts a posterior over a correction to the blind FFT+Rife estimate of that same trace, $q(\delta | \text{trace}, \log \sigma)$. The session context enters through a permutation-invariant attention block, so the network can pool as much or as little as the data support. Training uses instrument-anchored synthetic sessions (measured envelope population, measured noise law, blind fields) and evaluation is on the real traces; no real low-budget trace is used in training. Reported values are the median over 5 training seeds.

Table I and Fig. 1 show the outcome. The network does not reach the classical pooled fitter at the lowest budgets (504 nT at 5k, against 455 nT for shared-

envelope pooling and 743 nT for per-trace fitting). It is never itself the best estimator at any budget: pooled inference is ahead of it at every budget below 640k, by 3.5–21% (139 against 114 nT at 80k is the worst case, and the two are within 4% at 20k), and per-trace fitting is ahead at 640k (66 against 59 nT). What it does do is stay within 2–21% of the best per-budget classical estimator at *every* budget with a single set of weights, whereas each classical estimator is itself far off the best somewhere in the ladder — the pooled fitter by 23% at 640k and per-trace fitting by 63% at 5k. In other words the network has learned to stop sharing once sharing becomes harmful, which is the budget-dependent behaviour that the fitted law prescribes without the law being supplied to it. We stress that the network is *not* structure-free: it is conditioned on the noise level, trained on sessions with one shared phase frame and per-column relaxation times, and given the FFT estimate of each trace. What it is not given is the instrument’s sharing pattern, the crossover, or which parameters are common. It also does not yet match the classical pooled fitter in the regime where aggressive sharing is optimal, and we report that shortfall rather than presenting the network as uniformly superior. A controlled ablation in which the session-attention block is removed — so that each trace is estimated independently, with no cross-column information — is reported in Sec. IV G.

G. Ablation: removing the session context

To separate *learned sharing* from purely per-trace amortized inference we trained an identical architecture with the session-attention block removed, so that each trace is estimated independently; the training data, budget range and seed count are otherwise the same (5 seeds). Over the primary region the ablated model is worse at every budget except 20k, by 4–26% (559 against 504 nT at 5k, 145 against 139 nT at 80k, 83 against 66 nT at 640k), and the per-seed interquartile spreads of the two variants do not overlap except at 20k. Two conclusions follow, and we state them in the less flattering direction. First, the session context contributes, and contributes most at the highest budget, which is precisely where the classical pooled estimator is worst — consistent with the attention block learning to discount the shared envelope when the trace itself is informative. Second, the contribution is modest: the greater part of the network’s performance comes from the amortized residual inference itself, not from cross-column information. The network is therefore best described as a learned per-trace estimator with a weakly used session prior, not as a learned realisation of session pooling.

V. DISCUSSION

Three statements summarize the work.

(i) *At low photon budget, sensitivity is limited by nuisance identifiability rather than by shot noise or by optimisation.* Per-trace fitting is efficient against its own bound, so the room for improvement lies in the bound itself, and the bound can be moved by sharing the instrument’s nuisance parameters across the field settings of one sweep. In practice this is worth a factor $\sqrt{2.97}$ in precision, or a factor 2.97 in the number of repetitions needed to reach a target field uncertainty, whenever the budget is below r^* .

(ii) *The gain has a floor, and the floor is measurable instrument physics.* Sharing is only as good as the shared model. The relaxation envelope of our instrument varies by $\pm 5\%$ across the field range, and that variation sets a bias floor $b_{\text{pool}} \approx 57.6$ nT which is negligible at 5k repetitions and dominant at 640k. The two-term law of Eq. (2) together with the crossover $r^* \approx 62 \times 10^4$ repetitions is the quantitative version of the resulting rule: share while the photon term dominates, stop sharing when it no longer does. Because both terms are measurable before an experiment, A from the bound ratio and b from the envelope scatter, the crossover is predictable rather than merely empirical.

(iii) *A single amortized estimator reproduces the budget-dependent strategy.* A network trained once, across the whole 5k–640k range, tracks the better classical strategy at both ends of the ladder without being told the instrument model, the budget, or which parameters are shared. This is the practical value of the learning component: it recovers the budget-dependent strategy of a hand-specified hierarchical model without that model being hand-specified for this instrument, and it does so with one set of weights across the $128 \times$ budget range.

The law we report is *conditional* in two ways that we restate here rather than leaving to the appendix. It assumes an identifiability-filtered session (Table II and the add-back test of Table III quantify what happens otherwise), and it assumes that the blind FFT+Rife initialisation of each trace lands inside the refinement window — true for these traces, where the median initialisation error is 1.1×10^3 nT against a $\pm 3.2 \times 10^3$ nT window, but not guaranteed for another instrument or a different readout. The synthetic recovery tests we ran exercise the estimator and the initialisation under the assumed model; they do not by themselves certify the absolute field values of these real traces. Several further caveats belong in the same paragraph as these claims. First, all results are for one instrument, one diamond and one field range; the *form* of Eq. (2) is general, but b must be measured per instrument and per field range, since it is the envelope scatter that sets it. Second, the nominal applied field differs from the measured field by ~ 44 nT per column; above $\sim 3 \times 10^5$ repetitions that systematic, not the estimator, is the limiting factor for any method, and we mark that regime rather than claiming precision there. Third, the pooled estimators consume the whole session: our accounting charges each field estimate its own column’s accumulation time, which is the appropriate convention

when a swept-field session is the delivered product, and we report the single-column case, where pooling is unavailable, through the amortized estimator’s own performance. Fourth, columns 1–6 are excluded from the headline result: at low budget they are not identifiable from a single trace by any method we tried, and their calibrated treatment belongs to companion work [14].

The session-definition result of Table II sharpens all three statements into a two-step rule. The prescription is not “pool or do not pool” but *choose the session and the sharing pattern deliberately*. In the identifiability-filtered session the shared envelope costs little: full and partial pooling differ by 4.4 nT of floor (Table I), so sharing the relaxation time there is harmless. In the full sweep, by contrast, freeing the relaxation time removes 64 nT of floor (Table II and the text above). Applying that rule gives the scheduling consequence: if a target field uncertainty is specified, the cheapest protocol is the one whose floor lies below that target — pool aggressively while photons are scarce, relax the inhomogeneous parameters once the target approaches the floor, and stop adding repetitions when the floor is reached. Because the envelope scatter is measurable in a single high-budget calibration sweep, the whole decision can be made before the measurement.

VI. METHODS

Data. 8 sheets of 300 delays by 40 fields of DC Ramsey data on an NV ensemble; r from 5k to 640k repetitions per point; fields in steps of 1071.4 nT. Real traces are used for every reported metric, and the neural estimator is never trained on a real low-budget trace: its training set is generated synthetically from the measured envelope population and the measured noise law, and every performance number in Table I is evaluated on the real sheets.

Bounds. Fisher information computed numerically from Eq. (1), with $\partial s_i / \partial \theta$ by central differences. The per-trace free-envelope bound inverts the 6×6 Fisher matrix. The joint bound inverts the Fisher matrix of a 40-column session with a shared envelope, taking the field-diagonal block.

Classical estimators. Levenberg–Marquardt least squares (`scipy.optimize.least_squares`) with bounds, per-trace and pooled variants as described in the text. The pooled problems use each column’s FFT estimate as the initial field, so no method receives set-point information.

Amortized estimator. A spectral trunk (windowed rFFT magnitude, log-magnitude and unit-normalised real and imaginary parts of the same trace) followed by an MLP, with a permutation-invariant attention block over the session’s 40 columns and a per-column head that outputs a 512-bin posterior over the field correction δ relative to the trace’s own FFT+Rife estimate, spanning $\pm 8 \mu\text{T}$. Training minimises cross-entropy on instrument-

anchored synthetic sessions (envelope drawn from the measured population, per-column T_2^* , one shared phase frame per session, shot noise from the measured law, blind uniformly distributed fields), with fields restricted to positive values as in the experiment, for 30000 steps per seed. 5 seeds; reported values are the median over seeds.

Statistics. RMSE over columns with a bootstrap confidence interval (2000 resamples); signed bias; Wilcoxon signed-rank tests between paired estimators on the same traces, reported per budget. Multiplicity across the eight budget levels is handled by describing the ladder with Eq. (2) rather than as eight independent significance claims; the two lowest budgets form the pre-declared primary contrast.

Unit of replication. All measurements come from one swept-field session on one instrument, so the 34 columns of the primary region are not independent experimental replications: the column-to-column spread mixes genuine field-to-field variation with the shared per-column systematic of the magnet. The bootstrap intervals and Wilcoxon tests above therefore quantify how consistently a conclusion holds across field settings, not how it would generalise to other instruments or sessions. What we claim on this evidence is the existence and the location of the crossover on this instrument — a statement about estimator behaviour that is reproducible in simulation with exact ground truth (Sec. IV C) — and not a population-level effect size.

Provenance of the envelope prior. The envelope population of Sec. II is fitted from the highest-budget sheet, and it enters only two places: the reference CRLBs of Sec. III B and the synthetic training data of Sec. IV F. It is never used to fit a field, and every estimator reported in Table I fits its own nuisance parameters from the trace or session under test. In particular the per-trace and pooled estimators do not receive the envelope population, so the comparison between them is free of that prior.

DATA AND CODE AVAILABILITY

Code, result files and the scripts that generate every number in this paper are available at <https://github.com/hxm2023/nv-research4>. Laboratory raw data are available from the authors on reasonable request.

AI-USE DISCLOSURE

The analysis pipeline, the neural estimator and the manuscript drafting were developed with AI assistance under human supervision. All scientific claims were verified against the result files by the authors, who take responsibility for the content.

ACKNOWLEDGMENTS

To be completed.

APPENDIX: FIT AUDIT

Because the pooled fit is a large nonlinear problem, we audited it explicitly after an earlier version of this analysis was found to be corrupted by an optimiser artefact (a parameter-allocation bug that left 33 inactive parameters and stalled the solver; the subsection that reported it has been removed and its conclusion withdrawn).

For the 34-setting pooled model at every budget we record: the number of allocated parameters (39) and the number of active ones (39; the revised code allocates one relaxation-time parameter when it is shared and 40-many when it is free, so the inactive-parameter failure mode is structurally impossible); the agreement with an independent optimiser (L-BFGS-B) started from the same point, which returns identical fields to within 0.0 nT at all eight budgets; and the conditioning of the Jacobian. The raw condition number of $J^T J$ is 1.2×10^{11} at every budget, which is a statement about units — field parameters in nT against envelope parameters of order

unity — rather than about the problem: rescaling every column of J to unit norm gives a dimensionless condition number of 3.99–4.00, identical at all eight budgets. The fit is therefore well conditioned and contains no flat direction of the kind that produced the earlier artefact. The pooled solution is also stable against initialization: starting from the FFT estimate and from the per-trace maximum-likelihood solution converges to the same fields.

One limitation we do not hide: the whole pipeline is initialised from a blind FFT+Rife estimate, and its reliability is inherited from that initialization. On the real traces the FFT estimate is accurate to a median 1.1×10^3 nT, well inside the $\pm 3.2 \times 10^3$ nT window used by the refiners. Synthetic sessions drawn from the same envelope population have a comparable median initialisation error (0.9×10^3 nT over twelve sessions, range 0.7 – 1.2×10^3 nT), but a heavier tail: a minority of columns receive an initialisation that is far worse than the median, and because every estimator in the comparison shares that initialisation, those columns degrade all of them together. A synthetic-recovery test through the identical pipeline therefore exercises the initialisation as much as the estimator, and we report it as such rather than as a validation of the real-data numbers.

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